

**CENTER FOR DRUG EVALUATION AND
RESEARCH**

APPLICATION NUMBER:

208401Orig1s000

**CLINICAL PHARMACOLOGY AND
BIOPHARMACEUTICS REVIEW(S)**

CLINICAL PHARMACOLOGY REVIEW

NDA Number:	208401
Submission Type:	Original, 505 (b)(2) – Standard Review
Applicant Name:	Silvergate Pharmaceuticals Inc.
Submission Date:	June 30, 2015
Brand Name:	Qbrelis® (proposed)
Generic Name	Lisinopril
Dosage Form:	Oral Solution
Dosage Strengths:	1 mg/mL
Proposed Indication:	Treatment of hypertension in adults and children aged 6 years and older, adjunct therapy for heart failure in adults, treatment of acute (b) (4) in adults
OCP Division:	Division of Clinical Pharmacology 1
Primary Reviewer:	Rajanikanth Madabushi, PhD
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1.0 EXECUTIVE SUMMARY

In the current submission, Silvergate Pharmaceuticals, Inc. is seeking approval for a ready-to-use Lisinopril Oral Solution, 1 mg/mL, in accordance with 21 CFR 314.54 under the provisions of section 505 (b)(2) of the Food, Drug and Cosmetic Act. This application relies on the Agency's previous finding of safety and effectiveness for the reference listed drug, Zestril® (lisinopril) tablets (NDA 019777, approved May 19, 1988), distributed by AstraZeneca Pharmaceuticals.

Recently, enalapril for oral solution (Epaned®, NDA 204308) for the treatment of hypertension was approved in 2013 via a similar pathway.

Pursuant to section 526 of the Federal Food, Drug, and Cosmetic Act, the Office of Orphan Products Development granted Orphan Drug Designation request for Lisinopril Oral Solution, for the treatment of hypertension in pediatric patients 0 to 16 years of age. Based on orphan designation status the applicant is seeking waiver for pediatric studies in children (b) (4) years of age.

The primary focus of this review is the evaluation of bridging information between Lisinopril Oral Solution and Zestril®, the listed drug product.

1.1 Recommendations

Based on the submitted data demonstrating bioequivalence of lisinopril between oral solution and Zestril® tablets, the Office of Clinical Pharmacology recommends approval of lisinopril oral solution for the treatment of hypertension, adjunct treatment of (b) (4) heart failure and treatment of acute myocardial infarction in adults and treatment of hypertension in the pediatric population in the age between 6 and (b) (4).

1.2 Phase 4 Recommendations

None

1.3 Summary of Important Clinical Pharmacology and Biopharmaceutics Findings

Bioequivalence of Lisinopril with the Oral Solution and Zestril® Tablets

The sponsor submitted two reports of studies that investigated the exposure measures of lisinopril with an oral solution containing 10 mg lisinopril and a 10 mg Zestril® tablet in healthy subjects in the fasted and fed state. The 10 mg lisinopril dose corresponds to the initial dose of Zestril® recommended in the label. The geometric mean ratios of the exposure measures of C_{max} and AUC_{0-t} and the respective 90% confidence intervals in both studies were within the 80 to 125% bioequivalence limits indicating bioequivalence of the oral solution and 10 mg Zestril® tablet in the fasted and fed state.

2.0 QUESTION BASED REVIEW

The clinical pharmacology aspects of lisinopril have previously been described in detail for Zestril® under NDA 19777. Hence, an abbreviated version of the question based review is presented here to address the clinical pharmacology issues pertinent to this submission.

2.1 What are the proposed dosages and routes of administration?

The route of administration is oral. The proposed label for lisinopril oral solution and the label of Zestril® tablets (b) (4). For the treatment of hypertension in adults the recommended initial dose is 10 mg once-daily. The dosage should be adjusted according to blood pressure response. In pediatric patients in the age ≥ 6 years and $\text{GFR} > 30 \text{ mL/min/1.73m}^2$, the recommended starting dose is 0.07 mg/kg once daily (up to 5 mg once daily). Dosage should be adjusted according to blood pressure response up to a maximum of 0.61 mg/kg (up to 40 mg daily). For the treatment of heart failure, when used with diuretics (and usually digitalis) as adjunctive therapy for systolic heart failure, the initial dose is 5 mg once-daily. The recommended starting dose in these patients with hyponatremia (serum sodium $< 130 \text{ mL/min}$) is 2.5 mg once-daily and increased as tolerated to a maximum of 40 mg once-daily. For the reduction of mortality in acute myocardial heart infarction the recommended dose is 5 mg orally followed by 5 mg after 24 h, (b) (4) and then 10 mg once-daily. Dosing should continue for at least six weeks.

2.2 What are the design features of the clinical pharmacology and the clinical studies used to support dosing or claims?

The applicant performed two randomized crossover bioequivalence studies with lisinopril oral solution (10 mL x 1 mg/mL) and Zestril® tablets (10 mg) in healthy subjects in the fasted and fed state.

2.3 What is the relative bioavailability of lisinopril oral solution to Zestril® Tablets?

Following oral administration, lisinopril oral solution and Zestril® Tablets are bioequivalent. The geometric mean ratio for C_{max} was 0.94 (90% CI: 0.83 – 1.06) and $\text{AUC}_{0-\text{inf}}$ was 0.94 (90% CI: 0.86 – 1.02) under fasted conditions.

2.4 What is the effect of food on the bioavailability of the drug from the dosage form?

Following the ingestion of an FDA standard high calorie, high fat breakfast, the systemic exposure to lisinopril for both the products is decreased by 30-40% (cross-study comparison between fasted vs fed bioequivalence studies). Nevertheless, under fed condition, lisinopril oral solution and Zestril® Tablets are bioequivalent. The geometric mean ratio for C_{max} was 0.99 (90% CI: 0.96 – 1.03) and $\text{AUC}_{0-\text{inf}}$ was 0.99 (90% CI: 0.96 – 1.03). It should be noted that a minor negative food effect (10-20% decrease in lisinopril exposure) was also observed during the original NDA review for Zestril® tablets and deemed not clinically significant. The US Package Insert for Zestril® recommends dosing without regards to meals.

2.5. Analytical Section

Plasma concentration of lisinopril was measured by a validated LC-MS-MS method using lisinopril-D5 as internal standard. The bioanalytical methods satisfy the criteria for 'method validation' and 'application to routine analysis' set by the 'Guidance for Industry: Bioanalytical Method Development', and is acceptable. The validation aspects are discussed in detail in individual study review (GS03-01).

3.0 INDIVIDUAL STUDY REVIEWS

3.1 Fasted Bioequivalence Study (GS03-01)

Protocol: SG03-01	Study Period: 12/6-17, 2014
Title: A Randomized Single Dose, Two-Period, Two-Treatment, Two-Way Crossover Study to Determine the Relative Bioavailability of 10 mL of Lisinopril Oral Solution, 1 mg/mL, versus Zestril® 10 mg Tablets under Fasted Conditions in Healthy Adults	
Study Design: Open-label, randomized, two-period, two-treatment, cross-over study in which 56 healthy adult subjects received a single 10 mL dose of lisinopril oral solution, 1 mg/mL, in one period and a single Zestril Lisinopril 10 mg tablet in another period, following an overnight fast of at least 10 h	
☒ Bioequivalence	☐ Bioavailability
Background: Zestril (Lisinopril) is an oral long acting angiotensin-converting enzyme (ACE) inhibitor indicated for the treatment of hypertension in adult patients and pediatric patients 6 years of age and older to lower blood pressure. Zestril is also indicated to reduce signs and symptoms of systolic heart failure and for reduction of mortality in treatment of hemodynamically stable patients within 24 h of acute myocardial infarction. Following oral administration of Zestril, peak serum concentrations of lisinopril occur within about 7 h after administration. Food does not alter the bioavailability of lisinopril after administration of Zestril. Declining serum concentrations exhibit a prolonged terminal phase which does not contribute to drug accumulation. The terminal phase probably represents saturable binding to ACE and is not proportional to dose. Upon multiple dose administration lisinopril exhibits an effective half-life of 12 h. Lisinopril is not bound to plasma proteins and is excreted unchanged entirely in urine. Based on urinary recovery, the mean extent of absorption is approximately 25 (6-60) %. The absolute bioavailability of lisinopril is reduced to 16 % in patients with stable New York Heart Association (NYHA) Class II-IV congestive heart failure and the volume of distribution is slightly smaller than in healthy subjects. The absolute bioavailability of lisinopril in patients with acute myocardial infarction is similar to that in healthy subjects. Commonly reported adverse events after administration of Zestril include cough, dizziness, drowsiness, headache, depressed mood, nausea, vomitus, diarrhea, upset stomach, mild itching and rash.	
Rationale: There is currently no FDA approved, ready-to use oral liquid formulation of lisinopril for use by pediatric patients who require weight based dosing or older children and adults who have difficulty swallowing or cannot consume tablets. An oral solution of lisinopril would solve this problem.	
Objectives: To compare the rate of absorption and oral bioavailability of a test formulation of 10 mL of lisinopril oral solution, 1 mg/mL (Silvergate), to an equivalent oral dose of the commercially available comparator product, Zestril 10 mg tablet (Astra-Zeneca), when administered under fasted conditions. The 10 mg dose is the initial dose recommended by the label for Zestril tablets.	
Screening: Within 28 days prior to study start	Washout: ≥ 7 days

Periods: 2	Sequences: 2	
Treatments:		
	Test	Reference
Dosage Form	Lisinopril oral solution	Zestril tablets
Dosage Strength, mg	10 mL of 1 mg/mL	10 mg
Batch #	3126499	CH0066 distributed by AstraZeneca
Administration	oral	oral
Inclusion Criteria		
• Healthy male or female. Female were not to be pregnant or breast feeding in the age between 18 and 55 years of age (inclusive) and of body mass between 18-30 kg/m² and minimum weight of 50 kg • Female were to be postmenopausal at least 2 years before dosing or agreed to use an acceptable form of birth control from screening until 14 days after completion of the study, postmenopausal status to be confirmed with a follicle-stimulating hormone (FSH) test. Acceptable birth control included: Partner vasectomized, surgical or non-surgical sterilization, double barrier methods, abstinence, implanted or intrauterine hormonal contraceptives, oral, patch, or injected contraceptives, or vaginal hormonal devices • Vital signs: Measured after 3 min rest: Heart rate: 40-90 betas per min, systolic blood pressure: 90-140 mm Hg, diastolic blood pressure: 50-90 mm Hg		
Exclusion Criteria		
• History or presence of clinically significant cardiovascular, pulmonary, hepatic, renal, hematologic, gastrointestinal, endocrine, immunologic, dermatologic, neurologic, oncologic, or psychiatric disease or any other constitution that, in the opinion of the Investigator, would have jeopardized the safety of the subject or the validity of the results. • History of chronic cough, hyperkalemia, renal insufficiency, renal artery stenosis, or angioedema related to previous treatment with an ACE-inhibitor • Hereditary or idiopathic angioedema • Had significant abnormal finding on the physical exam, medical history, ECG, or clinical laboratory results at screening • History or presence of allergic or adverse response to lisinopril or related drugs, or any component of either study drug • Had used any over-the-counter (OTC) medication, including nutritional supplements, within 7 days prior the first dose of study medication • Had used any prescription drug medication, except hormonal contraceptives or hormonal replacement therapy within 14 days prior to the first dose of study medication • Had discontinued the use of implanted, intrauterine, or injected hormonal contraceptives within 6		

months prior to the first dose of study medication • Had discontinued the use of oral, patch, or vaginal hormonal contraceptives within 1 month prior to the first dose of study medication • Had been treated with any known drugs that are moderate or strong inhibitor/inducers of CYP enzymes such as barbiturates, phenothiazines, cimetidine, carbamazepine etc. within 30 days prior to the first dose of study medication and that in the Investigator's opinion may have impacted subject's safety or the validity of the study results • Had smoked or used any type of tobacco or nicotine based products (including e-cigarettes) within 60 days prior to the first dose of study medication • History of substance abuse or treatment (including alcohol) within past 2 years • Was a female with a positive pregnancy test result or breast feeding • Had a positive urine screen for drugs of abuse (amphetamines, barbiturates, benzodiazepines, cocaine, cannabinoids, opiates, alcohol, or cotinine) • Had a positive test result for hepatitis B surface antigen, hepatitis C antibody, or human immunodeficiency virus (HIV) at screening or had been previously treated for hepatitis B, hepatitis C, or HIV infection
Restrictions: • Subject could not consume salt substitutes • Beverages, and foods containing grapefruit or caffeine/xanthine (coffee, tea, cocoa, or chocolate) from 48 h prior to the first dose of study drug until the-end-of study visit • Subjects could not consume alcohol in any form from 7 days prior to the first dose of study drug through until the end-of-study visit • Subjects could not use tobacco or nicotine based products (including e-cigarettes) from 60 days prior to the first dose of study medication until the end-of-study visit • Subject could not engage in strenuous exercise from 48 h prior to the first dose of study drug until the end-of-study visit • If female was required to use one of the allowed forms of birth control listed in the inclusion criteria from screening until 14 days after completion of the study • If female could not discontinue the use of implanted, intrauterine, or injected hormonal contraceptives within 6 months prior to study start through the end-of-study visit • If female could not discontinue the use of oral, patch or vaginal hormonal contraceptives within 1 month prior to study start through the end-of study visit
Drug Administration: Study drugs were administered orally after a 10 h overnight fast together with 240 ml tap water of room temperature. The oral solution was administered via an oral dispensing syringe. The subjects were fasted for 4 h after drug administration
Dose Selection: The 10 mg doses selected for lisinopril was based consideration of the safety of the healthy subjects
Reserve Samples: Reserve samples were randomly selected and received from the inventory and retained by the site
Events Table:

Is shown below:

Table 9.5.1.1 Events schedule

PROCEDURE	SCREENING	STUDY	END-OF-STUDY
Informed consent	X		
Medical and medication histories	X	X	
ECG	X		X
Vital signs	X	X	X
Physical examination	X		X
Biochemistry, hematology, urinalysis	X		X
FSH (postmenopausal female subjects)	X		
Serology	X		
Urine drug, alcohol, and cotinine	X	X	
Pregnancy test (female subjects)	X	X	X
Drug administration		X	
Blood sample collection for pharmacokinetic analysis		X	
Adverse events		X	X

Sampling Times (PK, plasma)

- Pre-dose, 0.5,1,2,3,4,5,6,7,8,10,12,14,18,24,36,48,72 and 96 h after administration
- Pre-dose samples obtained from back-up subjects who were randomized into the study may have exceeded the pre-dose collection window.

Analytical Method: Within 60 min of collection the blood samples were centrifuged and the plasma samples frozen at -20 ° C. The plasma samples were received by (b) (4)

(b) (4) Duplicate samples also processed at the Clinical Site were received on (b) (4) All samples were frozen and in good condition. The characteristics of the assay used to analyze lisinopril in plasma and the performance of the assay during the analysis of the samples from Study SG03-01 are listed below:

Lisinopril	
Assay	LC-MS/MS with HPLC column, Internal standard:Lisinopril-D ₅
Matrix	Plasma
Calibration Range	0.545-328 ng/mL
Precision, %CV	<5.1 ^a
Accuracy %RE	-7.4- (-)-0.4 ^a , -3.1 – 11.1 ^b

^a QC samples during study SG-03-01 ^b Repeat samples during Study SG03-01

Validation	▪ Method validated prior to use	☒ Yes ☐ No
	▪ Method validation acceptable	☒ Yes ☐ No
Study Sample Analysis	▪ Samples analyzed within the established stability period	☒ Yes ☐ No
	▪ Quality control samples range acceptable	☒ Yes ☐ No
	▪ Chromatograms provided	☒ Yes ☐ No
	▪ Accuracy and precision of the calibration curve acceptable	☒ Yes ☐ No
	▪ Accuracy and precision of the quality control samples acceptable	☒ Yes ☐ No
	▪ Overall performance acceptable	☒ Yes ☐ No

Target PK Parameters: C_{max}, T_{max}, C_{last}, T_{last}, λ_z, t_{1/2}, AUC_{last}, AUC_{inf}, AUC_{Extrap} (%). The AUC_{last} was calculated using the linear trapezoidal rule, AUC_{inf} = AUC_{last} + C_{last}/λ_z

PK Analysis: Non-compartmental analysis in Phoenix WinNonlin (Version 6.3, Pharsight Corporation). In the pharmacokinetic analysis, below the limit of quantitation (BLQ) concentrations were treated as zero from time zero up to the time at which the first quantifiable concentration was observed; embedded and/or terminal BLQ concentrations were treated as “missing”. Actual sampling times were used for all pharmacokinetic and statistical analyses.

Sample Size: The sample size of 56 subjects was based on statistical considerations.

Statistical Methods: The natural logarithmic-transformed pharmacokinetic parameters were analyzed for differences between treatments using an ANOVA model with factors for sequence, subject within sequence and treatment. The log transformed C_{max} and AUC_{last} and AUC_{inf} were used in the statistical analysis as recommended in the FDA Guidance for establishing bioequivalence and the FDA Guidance for bioavailability and bioequivalence studies for orally administered drugs. Geometric means and 90% confidence intervals were determined for C_{max}, AUC_{last} and AUC_{inf}.

The Wilcoxon Rank test was used for a non-parametric comparison of T_{max}. A significant difference was defined a priori as p<0.05.

Changes in the Conduct of the Study: The letter of December 10, 2014 revised the dosing instructions for lisinopril oral solution, 1 mg/mL. (b) (4)

Results

Study Population :

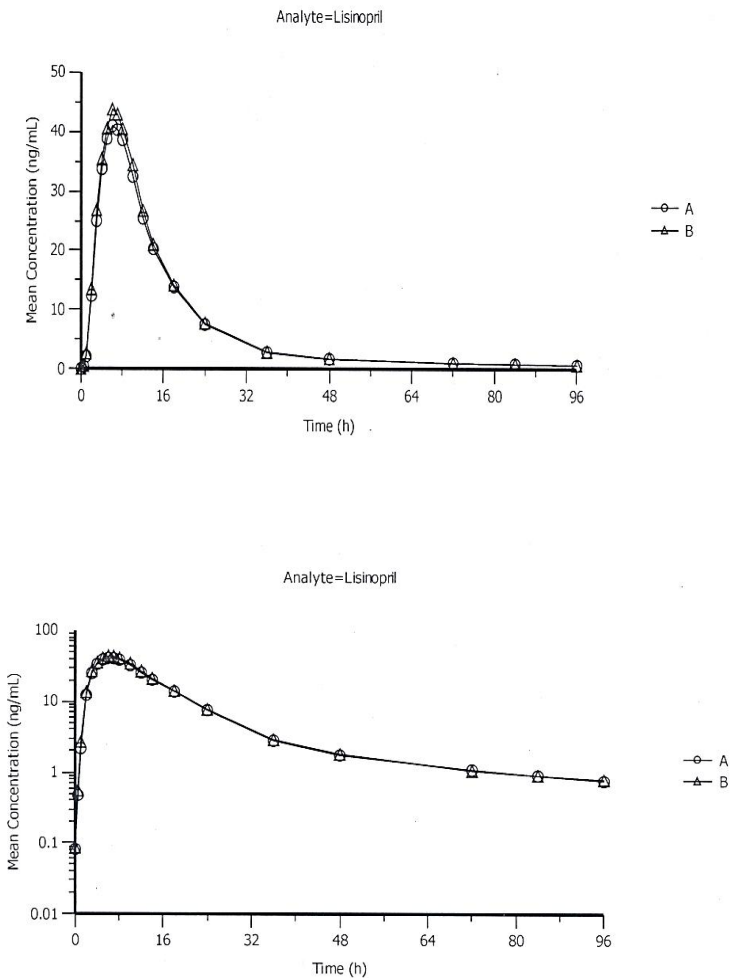
Randomized/Completed/Discontinued Due to AE	56/56/0
Age [Mean (range)], years	38 (19-55)
Weight [Mean (range), kg	72 (50-99)
Male/Female	19/37
Race (Caucasian/Black/Asian/Non-Hispanic or Latino)	46/7/3/24

PK

Quantifiable pre-dose concentrations for lisinopril were observed in 7 of the subjects. The concentrations in these pre-dose samples ranged between 0.582 and 0.918 ng/mL. Since they were < 5% of C_{max} of lisinopril the data for all subjects were included in the PK and statistical analyses without adjustment.

Linear and semi-logarithmic plots of the mean plasma concentrations after lisinopril 10 mg solution (Treatment A) and the Zestril 10 mg tablets (Treatment B) shown below indicate comparable plasma concentration profiles:

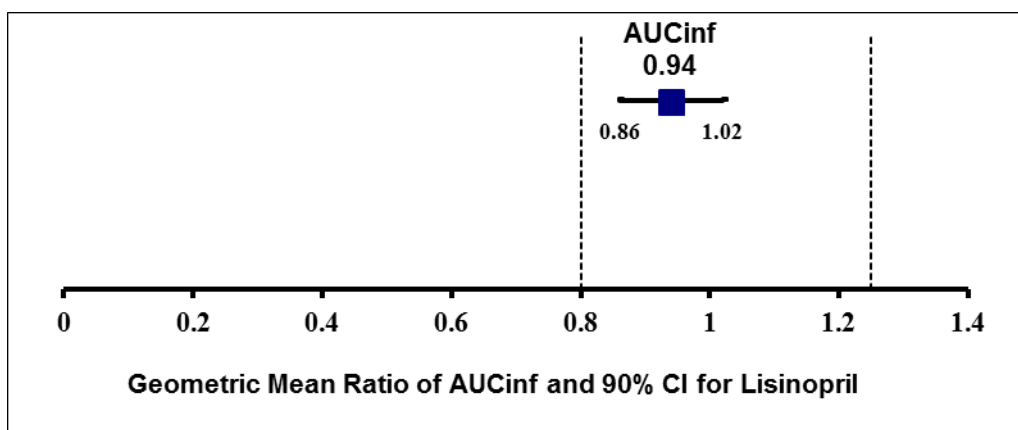
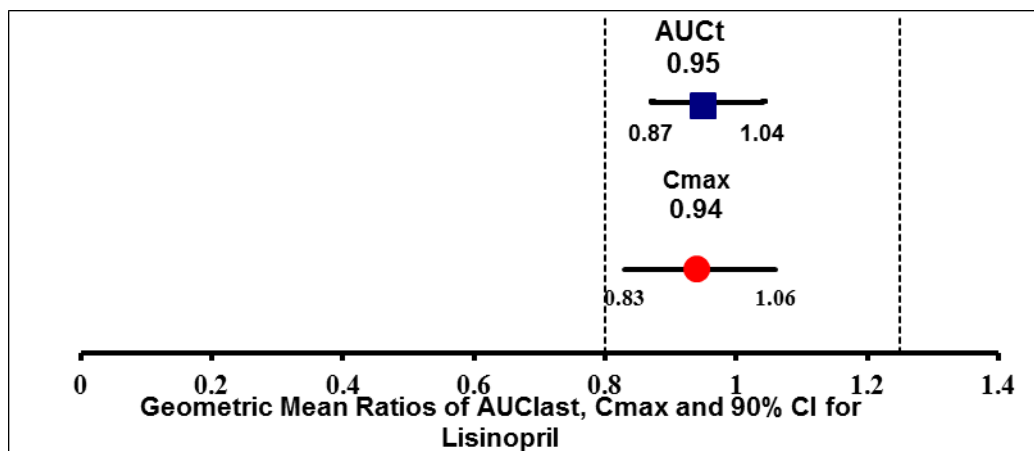
Figure 11.4.3.1 Mean Lisinopril Concentration-Time Profiles after Administration of the Test Formulation (Treatment A) and the Reference Product (Treatment B)



Treatment A = Test Formulation Lisinopril Oral Solution, 1 mg/mL
 Treatment B = Reference Product Zestril Lisinopril 10 mg Tablets
 Source data: Tables 14.2.1 and 14.2.2

In the large majority of subjects the plasma concentrations could be followed for 96 h after administration. The mean plasma concentration profiles after Lisinopril Solution and Zestril tablets are very similar.

The geometric means and 90% confidence intervals for the exposure parameters of lisinopril when comparing the solution relative to the Zestril tablet are shown in the below scheme:



The results indicate that rate and extent of lisinopril from the Lisinopril solution and the Zestril tablet are equivalent.

Mean (SD) t_{1/2} of the apparent terminal disposition phase for lisinopril after administration of the solution and the Zestril tablets was 46 (22) h and 55(38) h, respectively.

Site Inspected

Study Site: Requested: Yes ☒ No ☐ **Performed:** ☐ Yes No ☒ N/A ☐

Assay Lab: Requested: Yes ☒ No ☐ **Performed:** ☐ Yes No ☒ N/A ☐

On August 28, 2015 an OSIS Consult Request for Biopharmaceutical Inspections of the clinical and analytical sites was issued. On September 10 OSIS recommended to accept data without on-site inspection because a recent inspection of the clinical and analytical sites and was classified as No Action Required.

Safety

Was there any death or serious adverse events? Yes ☐ No ☒ NA ☐

Conclusion

The geometric means and 90% confidence intervals for the log transformed Cmax, AUClast and AUCinf were within the 80 to 125% bioequivalence limits.

Comments to be conveyed to the Applicant

- None

Appendix Method Validation

Preparation of Solutions

	Reference Standard	Internal Standard
Compound	Lisinopril Dihydrate*	[² H ₅]-Lisinopril
Source	The United States Pharmacopeial Convention, Inc.	(b) (4)
Lot / Batch	J0F043	TF-ALS-11-150-P1
Purity	100%	NA
Storage Condition	4 °C	4 °C
Expiration Date	Current Lot Verified	10 March 2016
Matrix	Human K ₂ -EDTA Plasma	
Source	(b) (4)	
Matrix	Human K ₂ -EDTA Lipemic Plasma	
Source	(b) (4)	
Matrix	Human K ₂ -EDTA Hemolyzed Plasma	
Source	(b) (4)	
Matrix	Human K ₂ -EDTA Whole Blood	
Source	(b) (4)	

*Certificate of analysis is not available.

QC Samples

Calibration standards were prepared at 0.545, 1.09, 2.72, 10.9 54.9, 164, 295 and 328 prepared on each day of each extraction and aliquotted at 0.200 mL.

The selected levels of the QC samples are shown in the below table:

QC/LLOQ Level	Concentration (ng/mL)	Date Prepared	Storage Temperature
Very High (Dilution)	1630	18 September 2014	-20 °C
High	261	23 September 2014	-20 °C
Medium	21.7	23 September 2014	-20 °C
Low	1.63	23 September 2014	-20 °C
LLOQ	0.545	18 September 2014	-20 °C

Assay Validation

Analytical runs consisted of at least 2 QCs at each level. Analytical runs were considered valid if at least (b) (4) % of the qualifying QC samples were within (b) (4) % of their theoretical values and at least (b) (4) % of the QCs at each level met this criterion. Validation runs consisted of 6 QCs at each level, including LLOQs.

Run #	Experiment
1	Precision and Accuracy, Dilution Integrity, Assay Selectivity, Detection Specificity, ICM
2	Reinjection Reproducibility
3	Precision and Accuracy, Recovery, Matrix Effect
4	Precision and Accuracy, BTS, FTS, XTS
5	Batch Size Determination, Partial Validation (Platform Addition Sciex 5000), Partial Validation (Platform Addition Multiplex)
6	Batch Size Determination, Partial Validation (Platform Addition Sciex 5000), Partial Validation (Platform Addition Multiplex)
7	Matrix Effect, WBS, LTS

BTS: bench top stability, FTS=freeze/thaw stability, XTS= extract stability, LTS= long-term stability

Standard Curves

The peak area ratios (y) of lisinopril to the internal standard and the concentration of the calibration standards (x) were fitted by weighted linear least squares regression analysis to the equation $y=a + bx$, where a is the y-intercept and b is the slope of the calibration curve. The assay procedure was found to be linear over the range of 0.545 to 328 ng/mL. The weighting of $1/x^2$ was determined based on power for weight calculations. All calibration standards were within 15.0% of their theoretical values, including the LLOQ standard. All 7 batches met this criterion.

LLOQ and OC Evaluation

Precision and accuracy at the LLOQ were evaluated by analyzing six samples at the lowest standard concentration of 0.545 ng/mL in each validation run. As shown in the below summary table the acceptance criteria, intra-run and inter-run precision and accuracy within 20% were met:

Analyte	Intrarun				Inter-run	
	CV		Bias		CV	Bias
	From	To	From	To		
Lisinopril	3.0%	6.8%	-3.5%	15.4%	8.8%	5.1%

Six samples from each QC level were evaluated in each validation run. The below summary table indicates that the QC samples met the acceptance criterion i.e. precision and accuracy within 15.0%:

Analyte	Intrarun				Inter-run			
	CV		Bias		CV		Bias	
	From	To	From	To	From	To	From	To
Lisinopril	0.9%	3.3%	-3.1%	6.9%	2.0%	3.7%	1.2%	5.4%

Dilution Integrity

Six replicates of the dilution QC level were diluted 10 fold prior to analysis. As shown by the below summary table dilution integrity is demonstrate by an intra-run and inter-run precision and accuracy within 15%.

Analyte	Intrarun	
	CV	Bias
Lisinopril	2.4%	4.9%

Stability Studies

Short Term benchtop and freeze/thaw stabilities in human plasma and extract stability in reconstituted form were evaluated at low and high QC levels with a freshly prepared calibration curve. QC samples were thawed and placed on the benchtop at room temperature for 24 h. A different set of samples was taken through five freeze/thaw cycles (-20 ° C to room temperature) A third set of samples was extracted and injected. The extracts were then re-injected after 95 h at 4 ° C using a freshly prepared calibration curve. The acceptance criteria applied were: At least ^(b)₍₄₎ % of the stability samples in each pool and the mean must be within ^(b)₍₄₎ % of their theoretical values. The accuracy ranged between -9.2 – 4.6%.

Reinjection Reproducibility

The extracts from an extractable run (Run 1) were stored at 4 °C. and then re-injected as Run 2. The acceptance

criteria, i.e. $\geq \frac{(b)}{(4)}\%$ of the QC samples were to be within the $\frac{(b)}{(4)}\%$ of their theoretical concentrations were met.

Recovery

Extraction recovery was evaluated in triplicate at low medium and high levels. The instrument response of pre-extraction spiked samples prepared in blank matrix (prepared by addition of analyte and internal standard (IS) to blank matrix prior to extraction) was compared to the instrument response of post-extracted spiked samples prepared in extracted blank matrix (prepared by addition of analyte and IS to blank matrix extract). For the evaluation of recovery the difference between the concentration levels and between the analyte and internal standard should be $\leq \frac{(b)}{(4)}\%$. Summary results are shown in the below table:

Analyte	From	To	Internal Standard	From	To
Lisinopril	69.52%	72.98%	Lisinopril-D ₅	66.48%	72.24%

Matrix Effect

The matrix effect, reported as matrix factor, was evaluated in six different (individual, non-pooled) lots, in singlicate, at high and low levels. In addition one hemolytic and one lipemic lot was evaluated in singlicate. The instrument response of post extraction spiked samples prepared in extracted blank matrix (prepared by addition of analyte and IS to blank matrix extract) was compared to the instrument response of matrix free spiked samples (prepared in the absence of blank matrix). For the evaluation of the matrix effect the matrix factor should be 0.800 to 1.20. The IS normalized matrix factor compares the analyte matrix factor to the IS matrix factor. The acceptance criterion to be met was that the precision of the IS-normalized matrix factor for the six different lots must be within 15.0%

The IS-normalized matrix factor for the lipemic lot at both levels and the hemolytic lot at the high level was between 0.80 and 1.20. The hemolytic lot at the lower level exceeded 1.20. The results are shown in the below 2 tables:

Table 9: Matrix Effect for Lisinopril and Internal Standard from Human K2-EDTA Plasma
in Study Masterfile 1004443 (Rm 3)

Lisinopril Peak Area					Lisinopril-D ₅ Peak Area			IS-Normalized Matrix Factor	% RCV
ng mL	Matrix Lot Number	Post-Extraction	Matrix-Free	Matrix Factor*	Post-Extraction	Matrix-Free	Matrix Factor*		
1.09	1354-72A	6101	5840	1.07	251306	228069	1.12	0.956	1.64
	1354-72B	6235	5318	1.09	251873	218539	1.12	0.975	
	1354-72C	6270	5925	1.10	249496	226232	1.11	0.990	
	1354-72D	6555		1.15	270282		1.21	0.955	
	1354-72E	6108		1.07	252303		1.13	0.953	
	1354-72F	6243		1.10	250335		1.12	0.982	
Mean Matrix Factor				1.10				0.969	
Therapeutic	1354-85	7155		1.26	328515		1.46	0.858	
295	1354-72A	1844886	1709019	1.08	256372	229983	1.11	0.979	1.85
	1354-72B	1832646	1695280	1.07	258568	233940	1.12	0.964	
	1354-72C	1870587	1710648	1.10	255231	231725	1.10	0.997	
	1354-72D	1950558		1.14	277489		1.20	0.956	
	1354-72E	1905791		1.12	260928		1.13	0.993	
	1354-72F	1880992		1.10	256072		1.10	0.999	
Mean Matrix Factor				1.10				0.981	
Hemolytic	1354-8C	2075568		1.22	294248		1.27	0.959	
Therapeutic	1354-85	2511828		1.47	338977		1.46	1.01	

*Matrix factor is calculated per matrix lot by dividing the post-extraction response by the mean of the matrix-free responses

Peak areas displayed as whole numbers

Note: Calculations performed with machine precision of Microsoft Office Excel

Table 9: Matrix Effect for Lisinopril and Internal Standard from Human K2-EDTA Plasma
in Study Masterfile 1004443 (Run 3)

Lisinopril Peak Area					Lisinopril-D ₅ Peak Area			IS-Normalized	
ng/mL	Matrix Lot Number	Post-Extraction	Matrix-Free	Matrix Factor*	Post-Extraction	Matrix-Free	Matrix Factor*	Matrix Factor	%CV
1.09	1354-72A	6101	5840	1.07	251306	228069	1.12	0.956	1.64
	1354-72B	6235	5318	1.09	251873	218539	1.12	0.975	
	1354-72C	6270	5925	1.10	249496	226232	1.11	0.990	
	1354-72D	6555		1.15	270282		1.21	0.955	
	1354-72E	6108		1.07	252503		1.13	0.953	
	1354-72F	6243		1.10	250335		1.12	0.982	
	Mean Matrix Factor			1.10			1.13	0.969	
Typemic	1354-85	7155		1.26	328535		1.46	0.858	
295	1354-72A	1844886	1709019	1.08	256372	229983	1.11	0.979	1.85
	1354-72B	1832646	1695280	1.07	258568	233940	1.12	0.964	
	1354-72C	1870587	1710648	1.10	255231	231725	1.10	0.997	
	1354-72D	1950558		1.14	277489		1.20	0.956	
	1354-72E	1905791		1.12	260928		1.13	0.993	
	1354-72F	1880992		1.10	256072		1.10	0.999	
	Mean Matrix Factor			1.10			1.12	0.981	
Hemolytic	1354-8C	2075568		1.22	294248		1.27	0.959	
Typemic	1354-85	2511828		1.47	338977		1.46	1.01	

*Matrix factor is calculated per matrix lot by dividing the post-extraction response by the mean of the matrix-free responses

Peak areas displayed as whole numbers

Note: Calculations performed with machine precision of Microsoft Office Excel

Assay Selectivity

Six different (individual, non-pooled) lots of blank plasma were fortified at the concentration of the LLOQ standard and extracted to examine the effect of variation in matrix. The acceptance criterion was that each lot must be within ^{(b) (4)}% of the theoretical value. The below summary table indicates that the lots met the acceptance criterion:

Analyte	Bias	
	From	To
Lisinopril	-16.9%	15.6%

Detection Specificity

Six different (individual, non-pooled) lots of blank plasma were analyzed to identify any interference at the retention time of lisinopril or the IS. The following acceptance criteria were met: If a peak was noted, the peak response at the retention time of the analyte and IS must be $\leq$ ^{(b) (4)}% and $\leq$ ^{(b) (4)}%, respectively, of the peak response of the LLOQ standard analyzed within the same run.

Interference of Common Medications

Two solutions containing approximately 10 µg/mL of common medications were tested for possible interference with lisinopril the IS. One solution contained acetaminophen, salicylic acid, brompheniramine, caffeine, cetirizine, chlorpheniramine, cimetidine, dextromethorphan, diphenhydramine, famotidine, ibuprofen, ketoprofen, loratidine, naproxen, omeprazole, pseudoephedrine, ranitidine, salicylic acid, and triprolidine () each. The second solution contained drospirenone, ethinyl estradiol, levonorgestrel, medroxyprogesterone acetate, norgestimate, and norethindrone. Each solution was injected in triplicate. The acceptance criteria were: For each solution tested, if a peak was noted, at least (b) (4) of the three replicates must have a peak response at the retention times of the analyte and IS $\leq$ (b) (4) % and $\geq$ (b) (4) %, respectively, of the peak response of the LLOQ standard analyzed within the same run.

Stability in Whole Blood

Whole blood samples were prepared at low and high concentrations of lisinopril. They were stored at room temperature for about 2 h and on ice. The samples were then centrifuged and stored frozen until analysis. The whole blood stability plasma samples were evaluated with a freshly prepared calibration curve and compared to time zero (To) samples that were immediately centrifuged after spiking. The acceptance criterion to be met included that the mean of the stability samples in each group must be within (b) (4) % of the samples of the To group. The below tables support stability of lisinopril in whole blood for 2 h at room temperature and on ice:

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Table 12: Whole Blood Stability for Lisinopril Extracted from Human K2-EDTA Plasma in Study 1004443

Hours	Low WBS T0 2.73 ng/mL	%Bias	Mean (%Bias)	Low WBS T120 Ice 2.73 ng/mL	%Bias	Mean (%Bias)	Low WBS T120 RT 2.73 ng/mL	%Bias	Mean (%Bias)
0	2.75	0.7	2.73 (0.0)						
0	2.72	-0.4							
0	2.67	-2.2							
0	2.79	2.2							
0	2.70	-1.1							
0	2.76	1.1							
2				2.80	2.6	2.76 (1.1)	2.77	1.5	2.76 (1.1)
2				2.73	0.0		2.79	2.2	
2				2.61	-4.4		2.65	-2.9	
2				2.75	0.7		2.77	1.5	
2				2.83	3.7		2.87	5.1	
2				2.82	3.3		2.70	-1.1	

Table 12: Whole Blood Stability for Lisinopril Extracted from Human K2-EDTA Plasma
in Study 1004443

Hours	Low WBS T0 2.73 ng/mL	%Bias	Mean (%Bias)	Low WBS T120 Ice 2.73 ng/mL	%Bias	Mean (%Bias)	Low WBS T120 RT 2.73 ng/mL	%Bias	Mean (%Bias)
0	2.75	0.7	2.73 (0.0)						
0	2.72	-0.4							
0	2.67	-2.2							
0	2.79	2.2							
0	2.70	-1.1							
0	2.76	1.1							
2				2.80	2.6	2.76 (1.1)	2.77	1.5	2.76 (1.1)
2				2.73	0.0		2.79	2.2	
2				2.61	-4.4		2.65	-2.9	
2				2.75	0.7		2.77	1.5	
2				2.83	3.7		2.87	5.1	
2				2.82	3.3		2.70	-1.1	

Long Term Stability

Long term stability of lisinopril in human plasma was evaluated at low and high QC levels with a freshly prepared calibration curve. QC samples were stored in propylene tubes at -20 °C for 15 days prior to analysis. The following acceptance criteria had to be met: $\geq$ (b) (4) % stability samples in each pool and the mean must be within (b) (4) % of their theoretical concentrations. The below Table indicates that the acceptance criteria were met:

Table 13: Long-Term Stability for Lisinopril Extracted from Human K2-EDTA Plasma
Storage Temperature = -20 °C
in Study 1004443

Days	QC High 261 ng/mL	%Bias	Mean (%Bias)	QC Low 1.63 ng/mL	%Bias	Mean (%Bias)
15	275	5.4	266 (1.9)	1.55	-4.9	1.56 (-4.3)
15	262	0.4		1.55	-4.9	
15	261	0.0		1.46	-10.4	
15	259	-0.8		1.63	0.0	
15	268	2.7		1.59	-2.5	
15	268	2.7		1.58	-3.1	

Stock Solution Stability

Stability of stock solution (lisinopril dehydrate in acetonitrile (1:1) at a concentration of 1.00 mg/ml was demonstrated for 72 days at -20 °C and at room temperature for 24 h. Stability of stock solution (lisinopril

dehydrate in acetonitrile (1:1) at a concentration of 6.55 µg/ml was demonstrated when in the refrigerator for 37 days and 24 h at room temperature.

Carryover

The matrix blank sample following the ULOQ standard is used to assess carryover. The acceptance criteria to be met included that if a peak was noted, at least (b) (4) of the three replicates must have a peak response at the retention times of the analyte and IS $\leq$ (b) (4) % and $\geq$ (b) (4) %, respectively, of the peak response of the LLOQ standard analyzed within the same run. The acceptance criteria were met.

Incurred Sample Reanalysis

Incurred sample reproducibility for lisinopril was evaluated by reanalyzing selecting samples near $C_{\max}$ and within the elimination phase for at least 10% of the samples across all subjects. IRS is demonstrated if at least two-thirds of the concentrations obtained for the original and repeat analyses do not deviate more than $\pm 20\%$ of their mean concentration. The results are shown in the below table (includes fasted and fed BE studies) and indicate that the acceptance criterion was met:

Samples Analyzed	Evaluable Results	Samples within 20%	Samples Outside of 20%	Percent Acceptable
n	n	n	n	%
224	224	210	14	94

3.2 Fed Bioequivalence Study (GS03-02)

Protocol: SG03-02	Study Period: 1/10-21,2015
Title: A Randomized Single Dose, Two-Period, Two-Treatment, Two-Way Crossover Study to Determine the Relative Bioavailability of 10 mL of Lisinopril Oral Solution, 1 mg/mL, versus Zestril® 10 mg Tablets under Fed Conditions in Healthy Adults	
Study Design: Open-label, randomized, two-period, two-treatment cross-over study in which 56 healthy adult subjects received a single 10 mL dose of Lisinopril oral solution, 1 mg/mL, in one period and a single Zestril Lisinopril 10 mg tablet in another period.	
☒ Bioequivalence	☐ Bioavailability
Background: Zestril (Lisinopril) is an oral long acting angiotensin-converting enzyme (ACE) inhibitor indicated for the treatment of hypertension in adult patients and pediatric patients 6 years of age and older to lower blood pressure. Zestril is also indicated to reduce signs and symptoms of systolic heart failure and for reduction of mortality in the treatment of hemodynamically stable patients within 24 h of acute myocardial infarction. Following oral administration of Zestril, peak serum concentrations of lisinopril occur within about 7 h after administration. Food does not alter the bioavailability of lisinopril after administration of Zestril. Declining serum concentrations exhibit a prolonged terminal phase which does not contribute to drug accumulation. The terminal phase probably represents saturable binding to ACE and is not proportional to dose. Upon multiple dose administration lisinopril exhibits an effective half-life of 12 h. Lisinopril is not bound to plasma proteins and is excreted unchanged entirely in urine. Based on urinary recovery, the mean extent of absorption is approximately 25 (6-60) %. The absolute bioavailability of lisinopril is reduced to 16 % in patients with stable New York Heart Association (NYHA) Class II-IV congestive heart failure and the volume of distribution is slightly smaller than in healthy subjects. The absolute bioavailability of lisinopril in patients with acute myocardial infarction is similar to that in healthy subjects. Commonly reported adverse events after administration of Zestril include cough, dizziness, drowsiness, headache, depressed mood, nausea, vomitus, diarrhea, upset stomach, mild itching and rash	
Rationale: There is currently no FDA approved, ready-to use oral liquid formulation of lisinopril for use by pediatric patients who require weight based dosing or older children and adults who have difficulty swallowing or cannot consume tablets. An oral solution of lisinopril would solve this problem.	
Objectives: To compare the rate of absorption and oral bioavailability of a test formulation of 10 mL of lisinopril oral solution, 1 mg/mL (Silvergate), to an equivalent oral dose of the commercially available comparator product, Zestril 10 mg (Astra-Zeneca) tablet, when administered under fed conditions. The 10 mg is the initial dose recommended in the Zestril label.	

Screening: Within 28 days prior to study start		Washout: ≥ 7 days	
Periods: 2		Sequences: 2	
Treatments:			
		Test	Reference
Dosage Form		Lisinopril oral solution	Zestril tablets
Dosage Strength, mg		10 mL of 1 mg/mL	10 mg
Batch #		3126499	CH0066 distributed by AstraZeneca
Administration		oral	oral
Inclusion Criteria			
• Healthy male or female in the age between 18 and 55 years of age (inclusive) and of body mass between 18-30 kg/m² and minimum weight of 50 kg • Females were not to be pregnant or breast feeding • Females were to be postmenopausal at least 2 years before dosing or agreed to use an acceptable form of birth control from screening until 14 days after completion of the study postmenopausal status to be confirmed with a follicle-stimulating hormone (FSH) test. Acceptable birth control included: partner vasectomized, surgical or non-surgical sterilization, double barrier methods, abstinence, implanted or intrauterine hormonal contraceptives, oral, patch, or injected contraceptives, or vaginal hormonal devices • Vital signs: Measured after 3 min rest: Heart rate 40-90 betas per min, systolic blood pressure 90-140 mm Hg, diastolic blood pressure: 50-90 mm Hg			
Exclusion Criteria			
• History or presence of clinically significant cardiovascular, pulmonary, hepatic, renal hematologic, gastrointestinal, endocrine, immunologic, dermatologic, neurologic, oncologic, or psychiatric disease or any other constitution that, in the opinion of the Investigator, would have jeopardized the safety of the subject or the validity of the results. • History of chronic cough, hyperkalemia, renal insufficiency, renal artery stenosis, or angioedema related to previous treatment with an ACE-inhibitor • Hereditary or idiopathic angioedema • Had significant abnormal finding on the physical exam, medical history, ECG, or clinical laboratory results at screening • History or presence of allergic or adverse response to lisinopril or related drugs, or any component of either study drug			

• Had used any over-the-counter (OTC) medication, including nutritional supplements, within 7 days prior the first dose of study medication • Had used any prescription drug medication, except hormonal contraceptive or hormonal replacement therapy within 14 days prior to the first dose of study medication • Had discontinued the use of implanted, intrauterine, or injected hormonal contraceptives within 6 months prior to the first dose of study medication • Had discontinued the use of oral, patch, or vaginal hormonal contraceptives within 1 month prior to the first dose of study medication • Had been treated with any known drugs that are moderate or strong inhibitor/inducers of CYP enzymes such as barbiturates, phenothiazines, cimetidine, carbamazepine etc. within 30 days prior to the first dose of study medication and that in the Investigator's opinion may have impacted subject's safety or the validity of the study results • Had smoked or used any type of tobacco or nicotine based products (including e-cigarettes) within 60 days prior to the first dose of study medication • History of substance abuse or treatment (including alcohol) within past 2 years • Was a female with a positive pregnancy test result or breast feeding • Had a positive urine screen for drugs of abuse (amphetamines, barbiturates, benzodiazepines, cocaine, cannabinoids, opiates, alcohol, or cotinine • Had a positive test result for hepatitis B surface antigen, hepatitis C antibody, or human immunodeficiency virus (HIV) at screening or had been previously treated for hepatitis B, hepatitis C, or HIV infection	Restrictions: • Subject could not consume salt substitutes • Beverages and foods containing grapefruit or caffeine/xanthine (coffee, tea, cocoa, or chocolate) from 48 h prior to the first dose of study drug until the end-of study visit • Subjects could not consume alcohol in any form from 7 days prior to the first dose of study drug through until the end-of-study visit • Subjects could not use tobacco or nicotine based products (including e-cigarette) from 60 days prior to the first dose of study medication until the end-of-study visit • Subject could not engage in strenuous exercise from 48 h prior to the first dose of study drug until the end-of-study visit • If female was required to use one of the allowed forms of birth control listed in the inclusion criteria from screening until 14 days after completion of the study • If female could not discontinue the use of implanted, intrauterine, or injected hormonal contraceptives within 6 months prior to study start through the end-of-study visit
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- If female could not discontinue the use of oral, patch or vaginal hormonal contraceptives within 1 month prior to study start through the-end-of study visit

Drug Administration: Study drugs were administered orally after a 10 h overnight fast followed by the ingestion of an FDA standard high-calorie, high fat breakfast meal. The subjects fasted for 4 h thereafter. Water was allowed ad lib during the study except for 1 h prior to dose administration through 1 h post-dose. The oral solution was administered via an oral dispensing syringe. After swallowing, the dose was followed with 240 mL water of room temperature. The subjects were fasted for 4 h after drug administration.

The FDA breakfast consisted of 2 eggs fried in butter, yolks broken, 2 slices of bacon, 4 oz. hash brown potatoes, 2 slices of white bread toast, 2 pats butter and 1 whole milk (8 oz)

Dose Selection: The 10 mg doses selected for lisinopril was based on consideration of the safety of the healthy subjects

Events Schedule: Is shown below:

Table 9.5.1.1 Events Schedule

PROCEDURE	SCREENING	STUDY	END-OF-STUDY
Informed consent	X		
Medical and medication histories	X	X	
ECG	X		X
Vital signs	X	X	X
Physical examination	X		X
Biochemistry, hematology, urinalysis	X		X
FSH (postmenopausal female subjects)	X		
Serology	X		
Urine drug, alcohol, and cotinine	X	X	
Pregnancy test (female subjects)	X	X	X
Standard high-fat, high-calorie breakfast		X	
Drug administration		X	
Blood sample collection for pharmacokinetic analysis		X	
Adverse events		X	X

Reserve Samples: Reserve samples were randomly selected and were received from the inventory and retained by the site

Sampling Times (PK, plasma)

- Pre-dose, 0.5,1,2,3,4,5,6,7,8,10,12,14,18,24,36,48,72 and 96 h after administration
- Pre-dose samples obtained from back-up subjects who were randomized into the study may have exceeded the pre-dose collection window.

Analytical Method: Within 60 min of collection the blood samples were centrifuged and the

plasma samples frozen at -20 ° C.

On January 21 and 22, 2015, the samples were received frozen and in good condition by (b) (4). The established long term stability of 91 days at -20 C ° was covering the period from sample collection to analysis. The plasma samples were analyzed for lisinopril using a validated LC-MS-MS procedure for a range from 0.545 to 328 ng/mL. The characteristics of the assay and its performance when the plasma samples containing lisinopril were analyzed are shown below:

Lisinopril	
Assay	LC-MS/MS with HPLC column Internal standard:Lisinopril-D ₅
Matrix	Plasma
Calibration Range	0.545-328 ng/mL
Precision, %CV	<8.7 ^a
Accuracy %RE	-7.3- 0.6 ^a

^a QC samples during Study SG03-02

Target PK Parameters: C_{max}, T_{max}, C_{last}, T_{last}, λ_z, t_{1/2}, AUC_{last}, AUC_{inf}, AUC_{Extrap} (%). The AUC_{last} was calculated using the linear trapezoidal rule, AUC_{inf}=AUC_{last} + C_{last}/λ_z

PK Analysis: Non-compartmental analysis in Phoenix WinNonlin (Version 6.3, Pharsight Corporation). In the pharmacokinetic analysis, below the limit of quantitation (BLQ) concentrations were treated as zero from time zero up to the time at which the first quantifiable concentration was observed; embedded and/or terminal BLQ concentrations were treated as “missing”. Actual sampling times were used for all pharmacokinetic and statistical analyses. Subjects that completed both study periods were included in the pharmacokinetic population.

Sample Size: The sample size of 56 subjects was based on statistical considerations.

Statistical Methods: The natural logarithmic-transformed pharmacokinetic parameters were analyzed for differences between treatments using an ANOVA model with factors for sequence, subject within sequence and treatment. The log transformed C_{max} and AUC_{last} and AUC_{inf} were used in the statistical analysis as recommended in the FDA Guidance for establishing bioequivalence and the FDA Guidance for bioavailability and bioequivalence studies for orally administered drugs. Geometric mean and 90% confidence intervals were determined for C_{max}, AUC_{last} and AUC_{inf}.

The Wilcoxon Rank test was used for a non-parametric comparison of Tmax. A significant difference was defined a priori as $p < 0.05$.

Changes in the Conduct of the Study: A revised protocol (Version 2.0) was issued on December 9, 2014 (b) (4)

(b) (4)

Results

Study Population :

Randomized/Completed/Discontinued Due to AE	56/53/3*
Age [Mean (range)], years	35 (19-53)
Weight [Mean (range), kg	72 (51-106)
Male/Female	32/24
Race (Caucasian/Black/Asian/Other	39/13/1/3

Two (2) subjects were withdrawn from the study prior to Period 2 due to protocol non-compliance. The first subject had a positive pregnancy test at Period 2 check-in and was to be contacted monthly and followed-up until the end of the pregnancy). The second subject did not complete breakfast in Period 2. One subject withdrew consent prior to Period 2. In accordance with the protocol no blood samples were analyzed from these 3 subjects. Thus, 52 subjects were the Pharmacokinetic population.

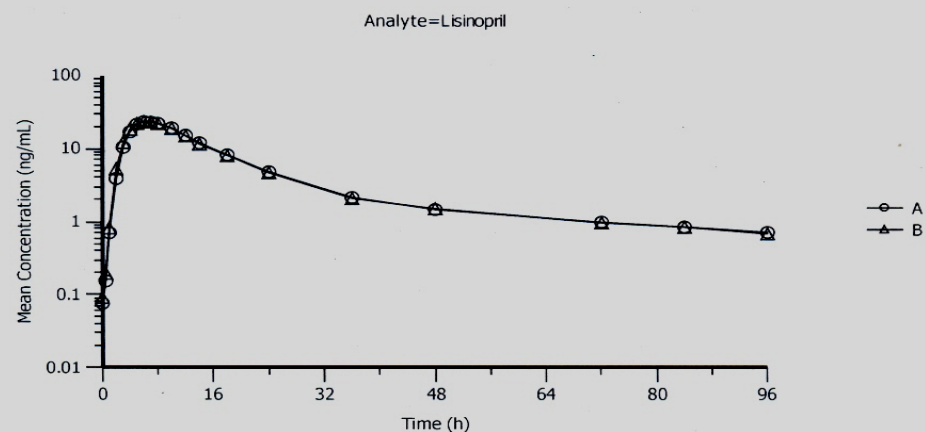
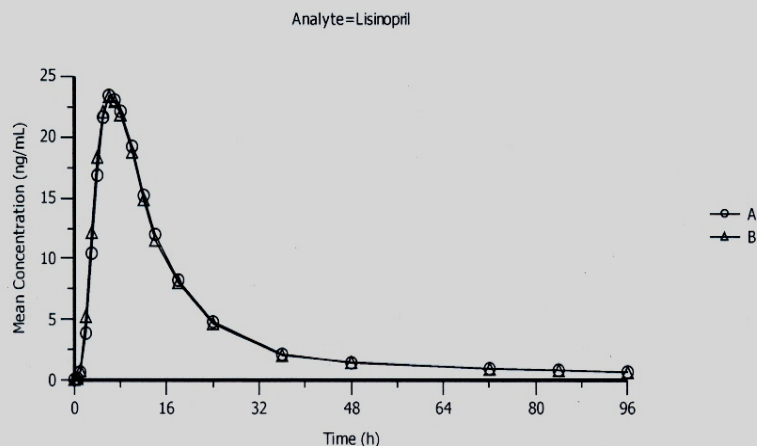
PK

Quantifiable pre-dose concentrations for lisinopril with Treatment A were observed in 7 of the subjects. The concentrations in these pre-dose samples ranged between 0.584 and 159 ng/mL. Quantifiable lisinopril concentrations in the pre-dose samples of Treatment B were observed in 8 subjects and ranged between 0.563 and 269 ng/mL. The large pre-dose concentrations of 159 and 269 ng/mL were observed in the same subject 305. Because they were $> 5\%$ of the corresponding Cmax values, Subject 305 was excluded from the pharmacokinetic population. In contrast, the quantifiable pre-dose concentrations in the other subjects were $< 5\%$ of the corresponding Cmax and

no adjustment was necessary.

Linear and semi-logarithmic plots of the mean plasma concentrations after lisinopril 10 mg solution (Treatment A) and the Zestril 10 mg tablets (Treatment B) shown below indicate comparable plasma concentration profiles:

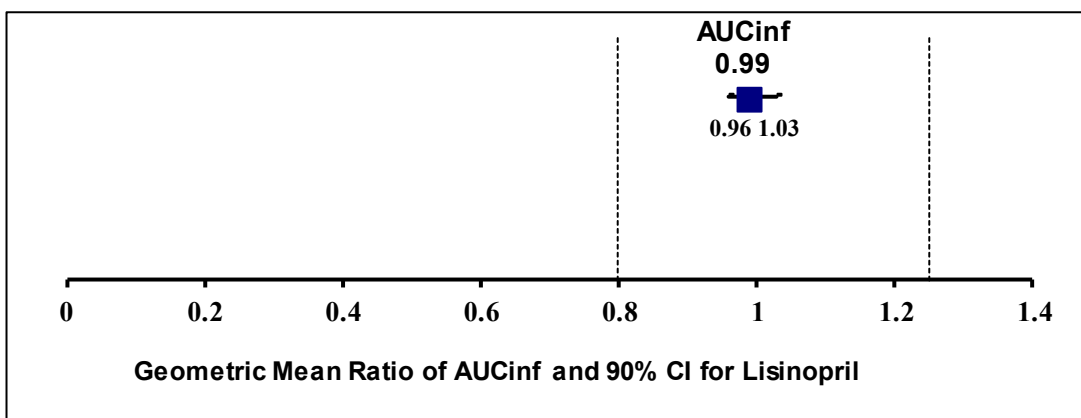
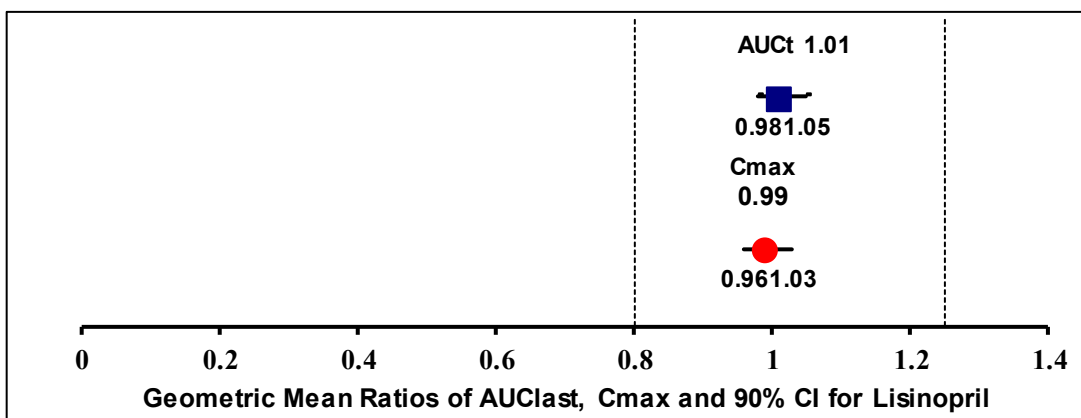
Figure 11.4.3.1 Mean Lisinopril Concentration-Time Profiles after Administration of the Test Formulation (Treatment A) and the Reference Product (Treatment B)



Treatment A = Test Formulation lisinopril oral solution, 1 mg/mL
Treatment B = Reference Product Zestril lisinopril 10 mg tablets
Source data: Tables 14.2.1 and 14.2.2

In the large majority of subjects the plasma concentrations could be followed for 96 h after administration. The mean plasma concentration profiles after administration of Lisinopril Solution and Zestril tablets appear to be very similar.

The geometric means and 90% confidence intervals for AUClast, Cmax and AUCinf of lisinopril when comparing the solution relative to the Zestril tablet are shown in the below scheme:



The mean (SD) $t_{1/2}$ of the apparent terminal disposition phase for lisinopril after administration of the solution and the Zestril tablets was 51 (25) h and 57(40) h, respectively.

Site Inspected

Study Site: Requested: Yes ☒ No ☐ Performed: Yes ☐ No ☒ N/A ☐

Assay Lab: Requested Yes ☒ No ☐ Performed: ☐ Yes No ☒ N/A ☐

On August 28, 2015 an OSIS Consult Request for Biopharmaceutic Inspections of the

clinical and analytical sites was issued. On September 10 OSIS recommended to accept data without on-site inspection because a recent inspection of the clinical and analytical sites and was classified as No Action Required.

Safety

▪ Was there any death or serious adverse events? Yes ☐ No ☒ NA ☐

Conclusion

The confidence intervals for the log transformed C_{max}, AUC_{last} and AUC_{inf} were within the 80 to 125 bioequivalence limits.

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/s/

RAJANIKANTH MADABUSHI
03/24/2016

SUDHARSHAN HARIHARAN
03/24/2016